Tony Wu

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Education

Massachusetts Institute of Technology — Cambridge, MA
B.S. in Mathematics — *Sep 2023 - May 2027 (expected)*
GPA: 4.72 on a 5.0 scale.

Honors and Selected Programs

International Mathematical Olympiad (IMO) - Gold Medal — Jul 2022
William Lowell Putnam Mathematical Competition - Rank 101 — 2023
MATS Fellow - Alignment Research Center stream — Summer 2026
MIT SPUR - Selected Participant — Summer 2025

Research Interests

Algorithmic statistics. Learning from dynamics and statistical–computational gaps in high dimensions.
Deep learning theory. How data structure and training dynamics shape learned representations.

Publications and Manuscripts

Learning Gaussian Graphical Models from a Glauber Trajectory Without Mixing.
Eric Shen*, **Tony Wu***, Mahbod Majid, and Ankur Moitra. **Equal first authors.*
ICML 2026, main conference. arXiv:2606.31230

Improved Rates of Learning Gaussian Graphical Models from Glauber Dynamics.
Tony Wu, Eric Shen, and Mahbod Majid.
Under review at ICLR 2027.

Research Experience

MIT Summer Program in Undergraduate Research (SPUR) — Jun 2025 - Aug 2025
Advisor: Ankur Moitra | Mentor: Mahbod Majid

- Co-developed a polynomial-time algorithm that recovers a sparse Gaussian conditional-independence graph from a single Glauber trajectory, with a trajectory-length guarantee independent of mixing time (ICML 2026; equal first author).
- Led the AI-assisted derivation of a time-t expansion of Glauber trajectories in subsequent follow-up work (ICLR 2027, under review).

Circuit Complexity Research — Sep 2025 - Present
Advisor: Daniel Reichman (WPI) | Mentor: Jiatu Li (MIT CSAIL)

- Used ideas from sigmoid computation to improve Amano's upper bound on the size of depth-two threshold circuits for inner product modulo two (IP_2).
- Studying with Jiatu Li what depth-two sigmoid circuits can compute without a prescribed output margin.

Alignment Research Center — Jun 2026 - Aug 2026
MATS Fellow and Visiting Researcher, Berkeley, CA | Mentor: Wilson Wu

- Designed output estimators for multilayer perceptrons with shifted weights. Used experiments to identify and remove redundant statistics from the estimators.
- Compared estimator performance with baseline methods and proved guarantees for specific cases of adversarial weight shifts.

Selected Coursework

Graduate statistics and machine learning. Deep Learning (6.7960, A+); Inference and Information (6.7800, A); Algorithms for Inference (6.7810, in progress); Information Theory: From Coding to Learning (6.7480, A); Mathematical Statistics: a Non-Asymptotic Approach (18.656, A).

Graduate mathematics. Commutative Algebra (18.705, A−); Algebraic Topology I (18.905, A); Number Theory I (18.785, B+).

Undergraduate mathematics. Real Analysis (introductory measure theory); Introduction to Topology; Algebraic Geometry (Harvard); Seminar in Number Theory.

Selected Projects

Local and Global Representation Alignment Fall 2025
Deep Learning (6.7960), MIT | Course project

- Collaborated on a study of local and global alignment across a broad set of word representations using CKA and CKNNA, motivated by the Platonic Representation Hypothesis.
- Observed persistent global differences and multiphase training dynamics before stabilization; discussed global representational rearrangement as a possible explanation in the final report.

Decoding Substitution Ciphers with a Change Point Spring 2025
Inference and Information (6.7800), MIT | Course project

- Built a substitution-cipher decoder using an n-gram Markov language model; combined simulated annealing with binary search to locate an unknown cipher-switch point.
- Placed in the top 15 on an individual decoding project in a class of 107 students.

Neural Network Interpretability and Steering Jul 2024 - Aug 2024
Participant, Program on AI and Reasoning (PAIR), FABRIC, Somerset, UK

- Experimented with steering vectors and prompt-injection defenses. Studied convolutional networks, transformers, and diffusion models.

Academic Service

International Mathematical Olympiad (IMO) 2025
Grader, Problem 6

Teaching and Outreach

Australian Olympiad Training Camps Dec 2022 - Present

- Support mathematics training for Australian students preparing for Olympiad competitions.

Global Teaching Lab Jan 2024
Instructor, Kigali, Rwanda

- Taught Olympiad mathematics to 40 local teachers and to secondary-school students; promoted the Rwandan mathematics program at six high schools.

Mathink May 2023 - Present
Main Course Designer and Teacher, New South Wales, Australia

- Design curricula and course materials that introduce students to university-level mathematics; deliver remote instruction in advanced mathematical topics.

Technical Skills

Python, PyTorch, NumPy, SciPy, C/C++, LaTeX, Git.